EYUEL TEFERA

Backend Engineer | Node.js, TypeScript, Java & Distributed Systems

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PROFESSIONAL SUMMARY

Backend Engineer with experience building and modernizing financial systems, payment services, and banking integrations. Strong background in Node.js, TypeScript, Java, PostgreSQL, MongoDB, Redis, and microservices architecture. Experienced in transaction processing, database design, API integrations, production troubleshooting, and backend performance optimization. Contributes to system architecture, code reviews, technical decisions, and the development of reliable financial services.

WORK EXPERIENCE

EDUCATION

Eaglelion System Technologies — Backend Engineer / Team Lead

Addis Ababa, Ethiopia | October 2024 – Present

Project: Dashen Bank Super App

Contribute to the backend modernization and development of a large fintech platform supporting payments, bank transfers, account services, and third-party integrations.

- Design, implement, and maintain backend services supporting financial transactions, account operations, payment processing, and external banking integrations.
- Contribute to the architecture and refactoring of services across more than 50 projects, improving consistency, maintainability, and adherence to engineering standards.
- Implement transaction-processing safeguards, including idempotency and duplicate-request handling, to improve reliability in financial workflows.
- Design and optimize database schemas, queries, indexes, and data-access patterns across PostgreSQL and MongoDB.
- Apply Redis caching and deduplication strategies where appropriate, balancing performance with data consistency and transaction correctness.
- Integrate banking and third-party systems for interbank transfers, foreign-currency transactions, utility payments, postpaid services, and card-related workflows.
- Improve asynchronous processing and notification workflows using Redis-backed queues and background jobs.
- Contribute to high-level and low-level technical designs, code reviews, and technical decisions while coordinating implementation and supporting other engineers.
- Investigate production issues using structured logging and observability tools, collaborating with QA teams to validate fixes and performance improvements.

Etech SC — Full Stack Developer / Backend Engineer (Contract)

Addis Ababa, Ethiopia | December 2023 – September 2024

Project: Savings and Credit Cooperative (SACCO) Management System

- Developed backend services for a financial management platform supporting savings, loans, budgeting, reconciliation, and financial reporting.
- Implemented financial calculation workflows for loan amortization, interest, penalties, dividends, and member-related financial operations.

- Designed REST APIs and PostgreSQL data models to support financial records, reporting, and business workflows.
- Developed background processing and notification workflows for computationally intensive operations, including dividend calculations.
- Contributed to database and tenant-management patterns supporting multiple SACCO organizations.
- Worked on reconciliation and financial reporting workflows, focusing on calculation accuracy and consistency.

Project: Share Management System

- Implemented dividend calculation and reinvestment workflows for member shares.
- Developed services for member registration, share tracking, and dividend distribution.
- Contributed to attendance tracking and stakeholder-management functionality.

Team Contributions

- Mentored junior developers in Java, Angular, and PostgreSQL.
- Conducted code reviews and collaborated on feature planning and implementation.
- Contributed to shared libraries and reusable components to improve consistency across projects.
- Participated in microservices design and continuous integration practices.

EDUCATION

Jimma University — Jimma Institute of Technology

Bachelor of Science in Computer Science
GPA: 3.32/4.00 · Exit Examination: 71/100

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Java, Go
- **Backend:** Node.js, Express.js, Java Vert.x, Spring Boot, Gin
- **Databases:** PostgreSQL, MongoDB, MySQL, Prisma ORM, SQL query optimization
- **Messaging & Caching:** Redis, BullMQ, background jobs, asynchronous processing
- **Architecture:** Microservices, REST APIs, event-driven systems, transaction reliability, idempotency
- **Infrastructure & Observability:** Docker, Kubernetes, GitLab CI/CD, Grafana, Loki, Pino
- **Engineering Practices:** API security, database schema design, code reviews, unit testing, production debugging, technical documentation

PERSONAL PROJECTS

API Gateway — Go

Technologies: Go, Gin, Fasthttp, JWT, Middleware, Unit Testing

- Designed a modular gateway architecture with JWT authentication, token-bucket rate limiting, and structured request logging.
- Established the foundation for reverse proxying and route-based load balancing.
- Developing automated tests for middleware and core gateway functionality.

SkillCraft — AI Agent Skills Compiler & CLI Engine *(Personal Project)*

Technologies: TypeScript, Node.js, Commander.js, Zod, Vitest, Architecture & Compiler Design

- Developed a production-grade CLI tool and multi-target compiler engine that transforms structured AI agent specifications (SkillSpec IR) into portable artifacts for Claude Code and Cursor.

- Engineered a robust Zod-based Abstract Syntax Tree (AST) schema and static linter enforcing trigger clarity, scope boundaries, and workflow actionability.
- Implemented multi-target transpilation with smart glob inference, atomic staging and installation pipelines, and static security auditing.
- Authored a comprehensive test suite using Vitest covering unit schema validation, authoring fixtures, compiler pipelines, and CLI smoke tests.

lazyALARM — Full-Stack IoT & Smart Alarm Platform *(In Active Development)*

Technologies: TypeScript, Node.js, Express, Prisma ORM, PostgreSQL, Zod, Vitest, OpenAPI/Swagger

- Designed and built a modular REST API backend for smart alarm scheduling, multi-device synchronization, and progressive wake-up routines.
- Architected a clean, layered backend structure (routes, controllers, domain services) enforcing strict resource ownership and data isolation across users and connected devices.
- Implemented robust runtime request validation using Zod and automated OpenAPI 3.0 documentation via Swagger UI.
- Authored a comprehensive automated test suite with Vitest and Supertest covering authentication, middleware, and core business domain logic.

SACCO SaaS — Multi-Tenant Core Banking & Financial Platform MVP *(Personal Project)*

Technologies: TypeScript, Node.js, Express, PostgreSQL (Row-Level Security), Redis, BullMQ, Docker, Zod, Vitest

- Architected a multi-tenant core banking engine for Savings & Credit Cooperatives (SACCOs), enforcing strict database-level data isolation using PostgreSQL Row-Level Security (RLS).
- Implemented an immutable double-entry accounting ledger featuring transaction atomicity, balanced debit/credit validation, and idempotency key deduplication.
- Built comprehensive Member Management and KYC modules supporting deterministic member numbering, document verification state machines, and audit logging.
- Developed the Savings and Shares subsystem handling account lifecycles, deposits, withdrawals, share transfers, and interest calculation workflows.
- Designed asynchronous job pipelines powered by Redis and BullMQ for scheduled interest compounding and batch ledger reconciliation, backed by an extensive integration test suite.

agent-hitl-telegram — Distributed Human-in-the-Loop Agent Authorization Engine *(Personal Project)*

Technologies: TypeScript, Node.js, Fastify, PostgreSQL, Prisma, Redis, BullMQ, Telegram Bot API (grammY), MCP

- Architected a distributed Human-in-the-Loop (HITL) authorization engine that intercepts privileged tool calls from AI agents (LangGraph, Vercel AI SDK, Claude Code, Cursor) and routes interactive approval requests to Telegram.
- Engineered atomic state transitions for approvals using Redis 7 and custom Lua scripts to prevent double-taps and race conditions, paired with an append-only PostgreSQL audit ledger.
- Designed robust asynchronous job pipelines via BullMQ for delayed TTL expirations, retries, and secure callback resumption with HMAC signature verification.
- Developed a Model Context Protocol (MCP) server enabling seamless integration across local and cloud-based AI environments with automatic argument redaction and data integrity hashing.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- AWS Certified Cloud Practitioner
- AWS Solutions Architect – Associate (In Progress)